

The empirical recurrence for OEIS A241075, recovered from its tail

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Abstract

OEIS A241075 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 104$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{96} = 432345564227567616 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A241075 is “Number of $n \times 4$ 0..2 arrays with no element equal to exactly one horizontal or vertical neighbor, with new values 0..2 introduced in row major order.”. It has offset 1 and begins

4, 39, 386, 4191, 47974, 572023, 6996566, 86779948, ...

The entry states:

Empirical recurrence of order 96 (see link above).

As of the “Last modified” line on the live entry (Jun 01 2026 15:03:57, revision 15) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 104$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 104$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 208$ exact terms from $n = 2$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	31	c_{25}	−51223829147301	c_{49}	−11008738631032792675	c_{73}	−3081458161958793
c_2	−358	c_{26}	92155801999439	c_{50}	−45514989717396178408	c_{74}	1530690632562302
c_3	2064	c_{27}	−252478043040840	c_{51}	10387073853438922992	c_{75}	2566615823973080
c_4	−7291	c_{28}	462311132038777	c_{52}	49899808278743064352	c_{76}	1623477355423561
c_5	19728	c_{29}	−752838650732416	c_{53}	141123212659398695008	c_{77}	6194608627632278
c_6	−57842	c_{30}	2189787455215948	c_{54}	277098100176468472576	c_{78}	2062348450580635
c_7	383913	c_{31}	−3183627088981387	c_{55}	197001906940630871168	c_{79}	4147402740102982
c_8	−2130755	c_{32}	4842017263398184	c_{56}	336084423404459714048	c_{80}	−2868174966894660
c_9	6511996	c_{33}	−5040463798498473	c_{57}	−134681346951192603648	c_{81}	201801167933244
c_{10}	−18365374	c_{34}	7146901550872165	c_{58}	−339459418917220976128	c_{82}	−2685036481131636
c_{11}	49611322	c_{35}	−14952995041632790	c_{59}	−1303148598897698628608	c_{83}	−2231453932740803
c_{12}	−124487723	c_{36}	−17833999580566604	c_{60}	−819409380174653833216	c_{84}	9044269750035551
c_{13}	466350081	c_{37}	14192143319792654	c_{61}	−1826287108301708693504	c_{85}	4740370298854587
c_{14}	−631134150	c_{38}	−159439180806420437	c_{62}	−444496574592426770432	c_{86}	1298221525038404
c_{15}	−44155659	c_{39}	249639432396215467	c_{63}	96531174454260596736	c_{87}	1695256909682300
c_{16}	−2611159119	c_{40}	−466140191204401407	c_{64}	2375212488287782240256	c_{88}	5134315473083270
c_{17}	−2659657073	c_{41}	1252854391112290655	c_{65}	2765626166248211939328	c_{89}	−681920154994860
c_{18}	14525178169	c_{42}	−868915303862908281	c_{66}	3232891351480119328768	c_{90}	−363192721580549
c_{19}	−111250579258	c_{43}	4095045807594448502	c_{67}	3342123447121854857216	c_{91}	2378771768304541
c_{20}	621723915903	c_{44}	−2782655667505399790	c_{68}	1282246902305368047616	c_{92}	−517519892180212
c_{21}	−773857553426	c_{45}	6192725123063256379	c_{69}	−2774360316401829281792	c_{93}	6438120847307492
c_{22}	1618457145129	c_{46}	−9817955405481033040	c_{70}	−2720552633843211829248	c_{94}	−128712877350248
c_{23}	−8240033871986	c_{47}	−1316763696946479844	c_{71}	−4145710327287909449728	c_{95}	−38730956795386
c_{24}	22674209907324	c_{48}	−32964413629067307594	c_{72}	−5600259464775770046464	c_{96}	432345564227567

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A241075 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{96-1} - \dots - c_{96}$. Its constant term is $-c_{96} = -432345564227567616$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 432345564227567616.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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